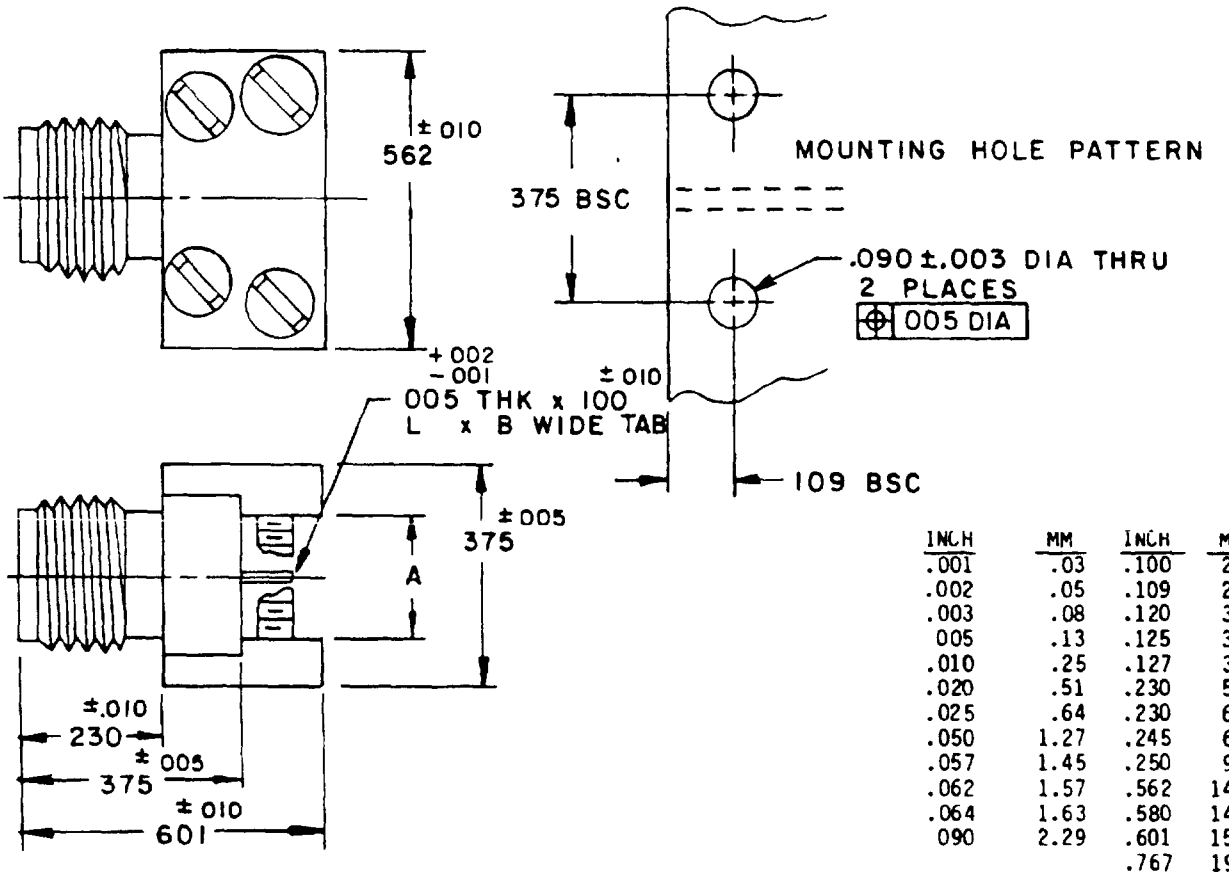
MILITARY SPECIFICATION SHEET

CONNECTOR, COAXIAL, RADIO FREQUENCY, STRIP OR MICROSTRIP TRANSMISSION LINE, SERIES SMA (SOCKET CONTACT, END LAUNCH RECEPTACLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-83517.



INCH	MM	INCH	MM
.001	.03	.100	2.54
.002	.05	.109	2.77
.003	.08	.120	3.05
.005	.13	.125	3.18
.010	.25	.127	3.23
.020	.51	.230	5.84
.025	.64	.230	6.22
.050	1.27	.245	6.35
.057	1.45	.250	9.53
.062	1.57	.562	14.27
.064	1.63	.580	14 73
.090	2.29	.601	15.27
		.767	19.48

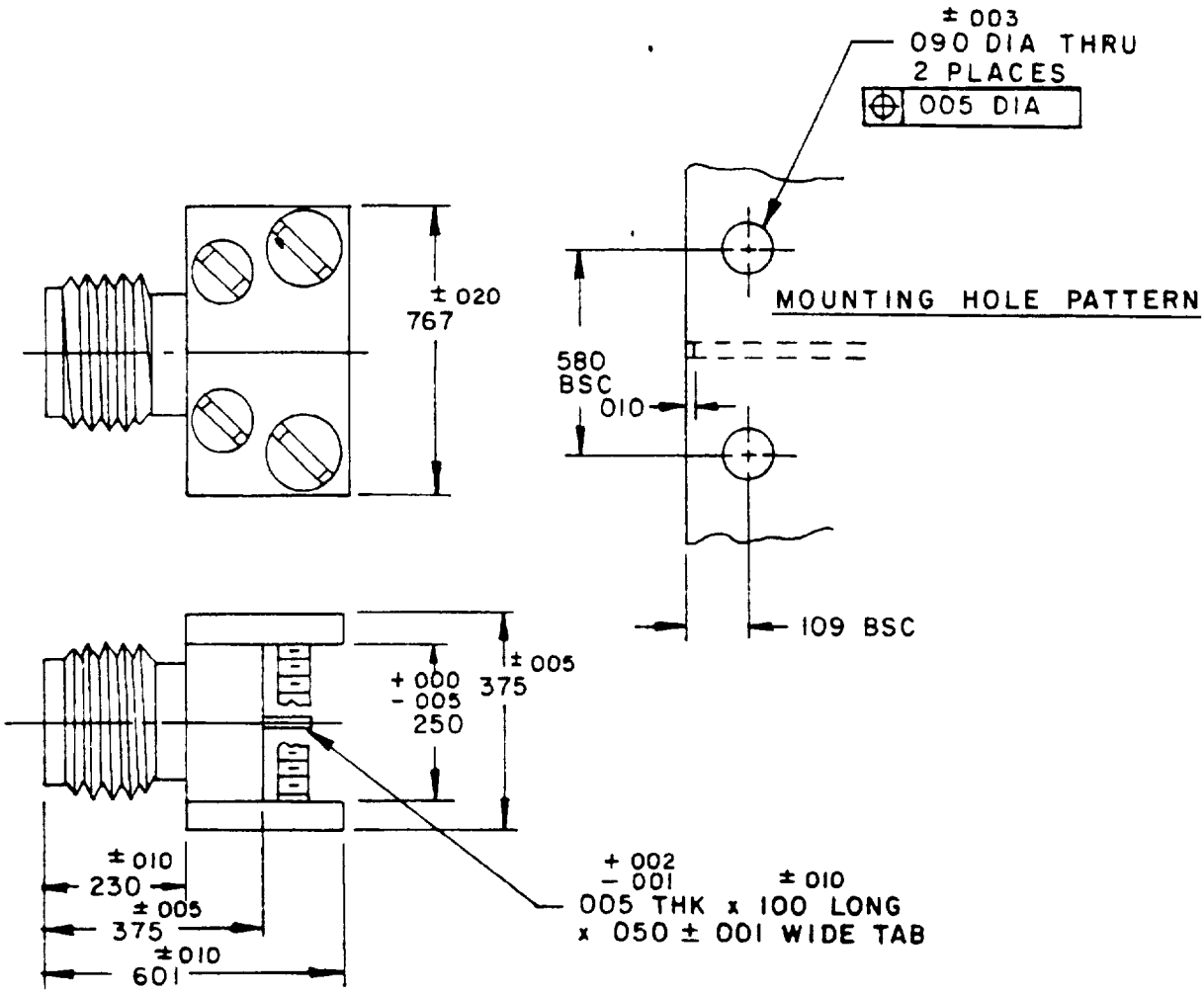
PART NO.	"A"DIM	"B"DIM	FOR TRANSMISSION LINE THICKNESS
-31001	.064/.057	.025±.002	.0625
-31002	.127/.120	.050±.001	.125
-31003	250/.245	.050±.001	.125 WITH(2).063 SUPPORT PLATES

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25 4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1 Series SMA, socket contact, end launch, receptacle

FSC 5935

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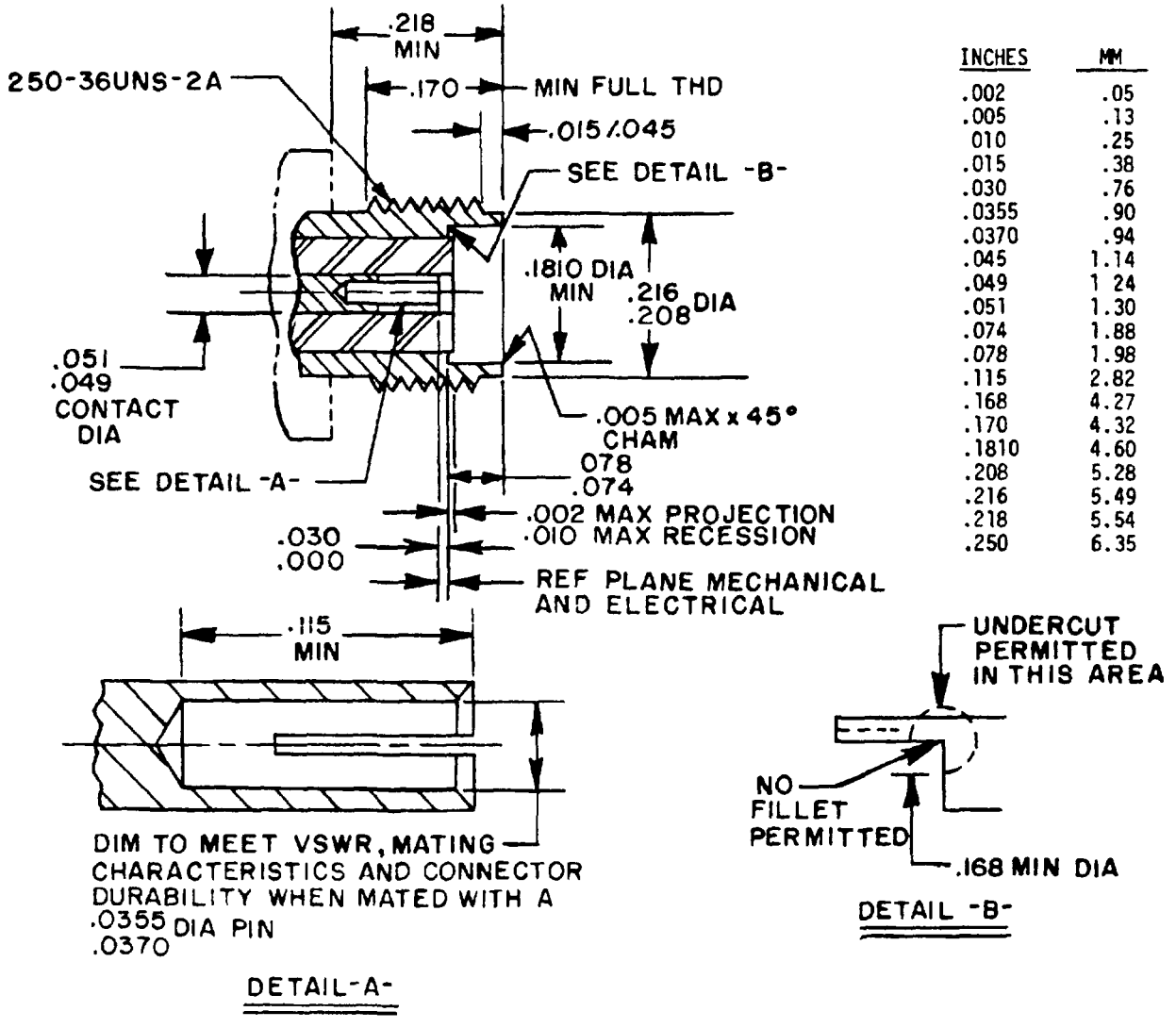
INCH	MM	INCH	MM
.001	.03	100	2.54
.002	.05	.109	2.77
003	08	120	3 05
005	.13	.125	3.18
.010	.25	.127	3 23
020	51	230	5 84
025	64	230	6 22
050	1 27	.245	6 35
057	1 45	250	9.53
062	1 57	.562	14.27
064	1 63	580	14 73
090	2 29	601	15 27
		767	19 48

PART NO.	TRANSMISSION LINE THICKNESS
-31004	.250

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm
3. All undimensioned pictorial configurations are for reference purposes only

FIGURE 2. Series SMA socket contact end launch, receptacle (.25 inch)

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INCHES	MM
.002	.05
.005	.13
010	.25
.015	.38
.030	.76
.0355	.90
.0370	.94
.045	1.14
.049	1 24
.051	1.30
.074	1.88
.078	1.98
.115	2.82
.168	4.27
.170	4.32
.1810	4.60
.208	5.28
.216	5.49
.218	5.54
.250	6.35

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm
3. All undimensioned pictorial configurations are for reference purposes only

FIGURE 3 Mating dimensions for socket termination

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ENGINEERING PARAMETERS

Nominal impedance 50 ohms

Voltage rating 600 Vrms maximum at sea level.

150 Vrms maximum at 70,000 feet.

Frequency range 0 to 18.0 GHz.

Temperature rating -65° to 105°C.

REQUIREMENTS

Design and construction See figures 1 through 3 and table I.

Force to engage and disengage

Torque - 2 inch-pounds maximum.

Longitudinal force - Not applicable.

Coupling proof torque Not applicable.

Inspection note: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Contact gaging: See figure 4.

Contacts with spring members

Center contact (socket)

Oversize test pin - .0375 +.0001

Test pin finish - 16 microinches.

Insertion depth - .030/.045.

Number of insertions - 3.

Insertion force test Steel test pin diameter .0370 +.0001.

Insertion depth - .050/.075.

Test pin finish - 16 microinches.

Insertion force - 3 pounds maximum.

Withdrawal force test. Steel test pin diameter 0355 - .0001.

Insertion depth - .050/.075.

Withdrawal force - 1 ounce minimum.

Test pin finish - 16 microinches

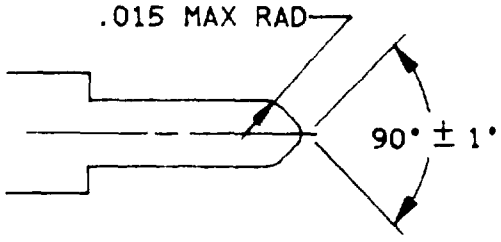


FIGURE 4 Test pin data

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TABLE I. Part number and characteristics.

Part no. M83517/09-	Connector figure no.	Characteristics
31001	1	Strip transmission line end launch, receptacle .025 wide tab, for 1/16" strip transmission line.
31002	1	Strip transmission line end launch, 050 receptacle .050 wide tab, for 1/8" strip transmission line.
31003	1	Strip transmission line end launch, receptacle, .050 wide tab, (2).063 support plates, for 1/8" strip transmission line.
31004	2	Strip transmission line end launch, receptacle, .050 wide tab, for 1/4" strip transmission line.

Permeability of nonmagnetic materials. Applicable.

Seal

Hermetic sealed connectors: Not applicable.

Pressurized and weatherproof connectors: Not applicable.

Insulation resistance 5,000 megohms minimum.

Center contact retention.

Axial force: 6 pounds minimum.

Torque 4 inch-ounces minimum.

Dielectric withstanding voltage Applicable, test condition 1.

Test voltage 1,000 Vrms.

Corrosion Applicable, test condition B.

Voltage standing wave ratio (VSWR)

Test frequency range From 5 to 18.0 GHz.

Swept frequency VSWR test setup

Step 1 See basic specification.

Step 2 VSWR shall be less than 1.080 + 005 frequency (frequency in GHz).

Item 11p VSWR shall be less than 1.025 + .002 frequency (frequency in GHz).

Item 11j VSWR shall be less than 1.025 + .002 frequency (frequency in GHz)

Step 3 VSWR shall be less than (to be determined).

Test fixture - See figure 1 of basic specification.

Step 4 VSWR test shall be less than (to be determined).

RF transmission loss (To be determined)

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RF leakage. Not applicable.

Connector durability

Interface

500 cycles minimum at 12 cycles/minute maximum rate.

Connector shall meet contact gaging and force to engage and disenyage requirements.

Contact resistance In milliohms maximum

	Initial	After environment
Center contact	3.0	4.0
Outer contact	2.0	Not applicable

Thermal shock: Applicable, test condition A.

Moisture resistance Method 106 of MIL-STD-202.

No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity

RF high potential withstanding voltage

At a frequency between 5 to 7.5 MHz

Leakage current - Not applicable.

RF voltage - 1,000 Vrms.

Coupling mechanism retention force Not applicable.

Part number M83517/9-(dash number from table 1.)

Group qualification See table II.

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors 1/2/	Qualifies the following connectors
I	3+001	3+001 3+002 3+003 3+004

1/ Individual connectors other than listed are self qualifying only

2/ Qualification of connectors qualifies connectors of the same material only.

\+ Denotes finish.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - AR, MI, AT
Navy - SH, AS
Air Force - 11, 99
DLA - ES

User activities
Navy - MC
Air Force - 19

Preparing activity.
Air Force - 85

(Project 5935-3159-9)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-83517/9 - Connector, Coaxial

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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